Pandas Data Analysis Project

Project Objective

The objective of this project is to perform data analysis using Python and Pandas library. The project demonstrates loading CSV data, cleaning data, filtering records, grouping data, and generating insights.

Technologies Used

- Python
- Pandas
- Matplotlib
- VS Code

Dataset Used

Supermarket Sales Dataset (CSV format)

Features Implemented

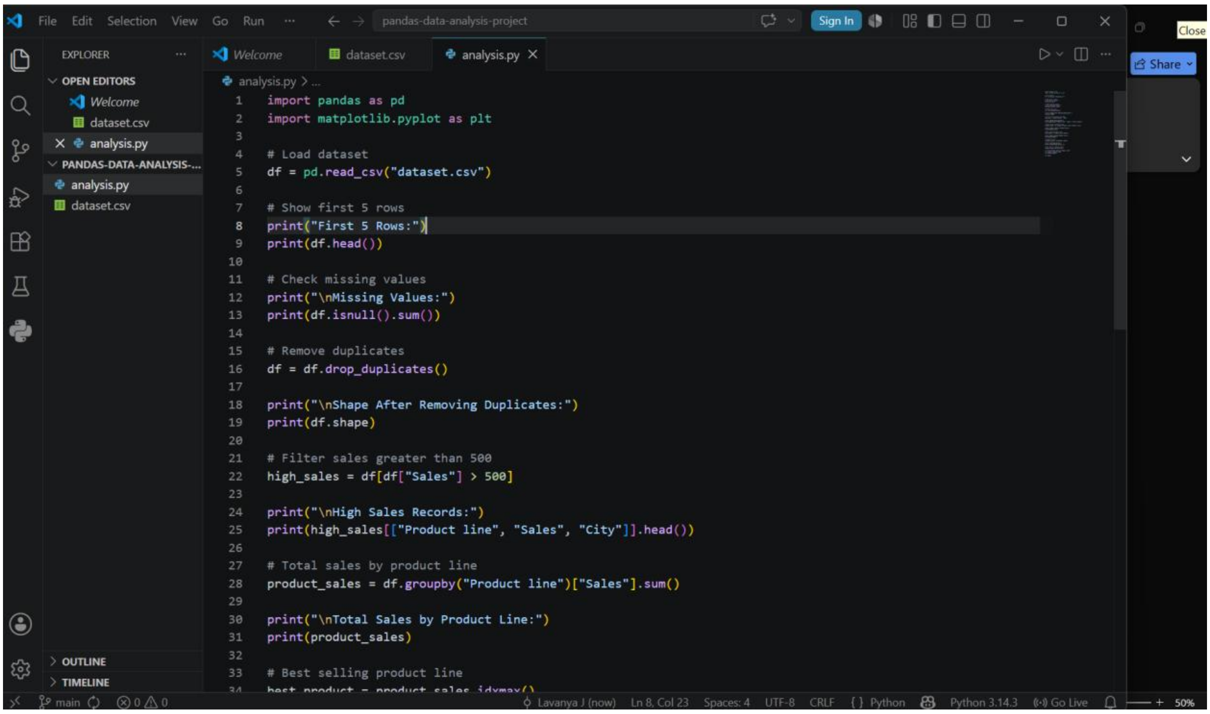
- Data Loading using Pandas
- Data Cleaning
- Missing Value Checking
- Filtering Data
- Grouping and Aggregation
- Data Visualization using Bar Chart
- Insight Generation

Code Snippet

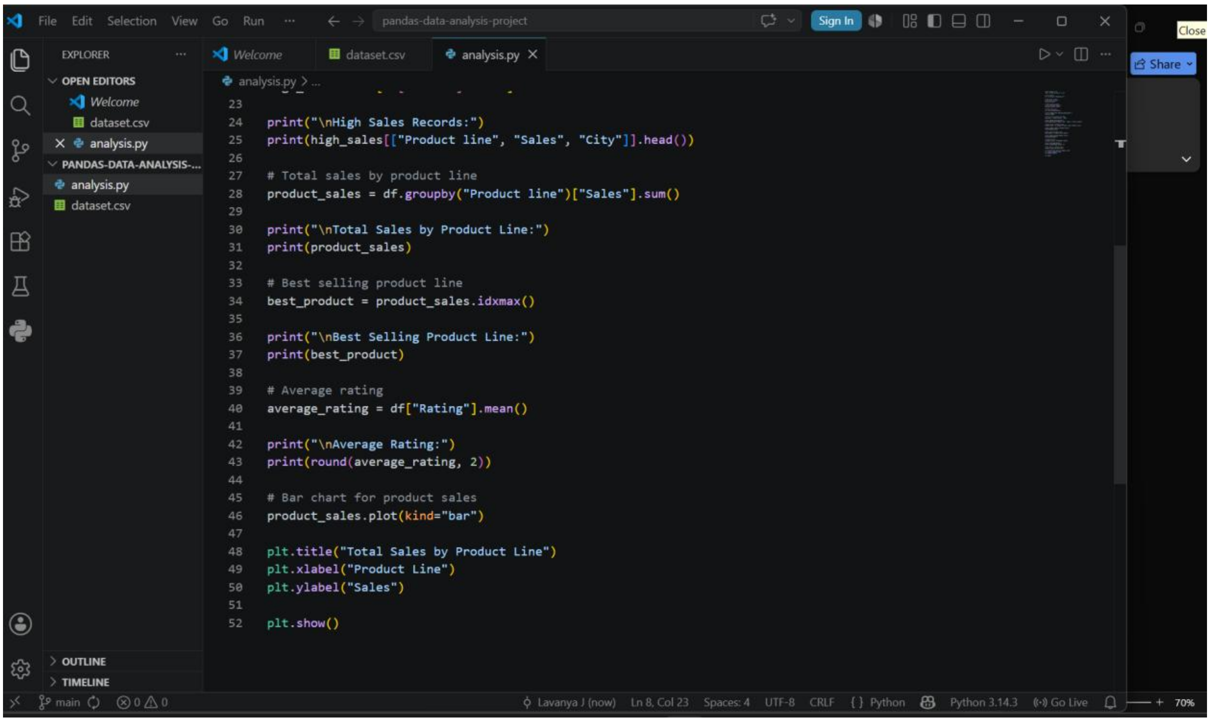
```
import pandas as pd

import matplotlib.pyplot as plt


df = pd.read_csv("dataset.csv")
```

```
product_sales = df.groupby("Product line")["Sales"].sum()

product_sales.plot(kind="bar")
plt.show()
```

Code Screenshots

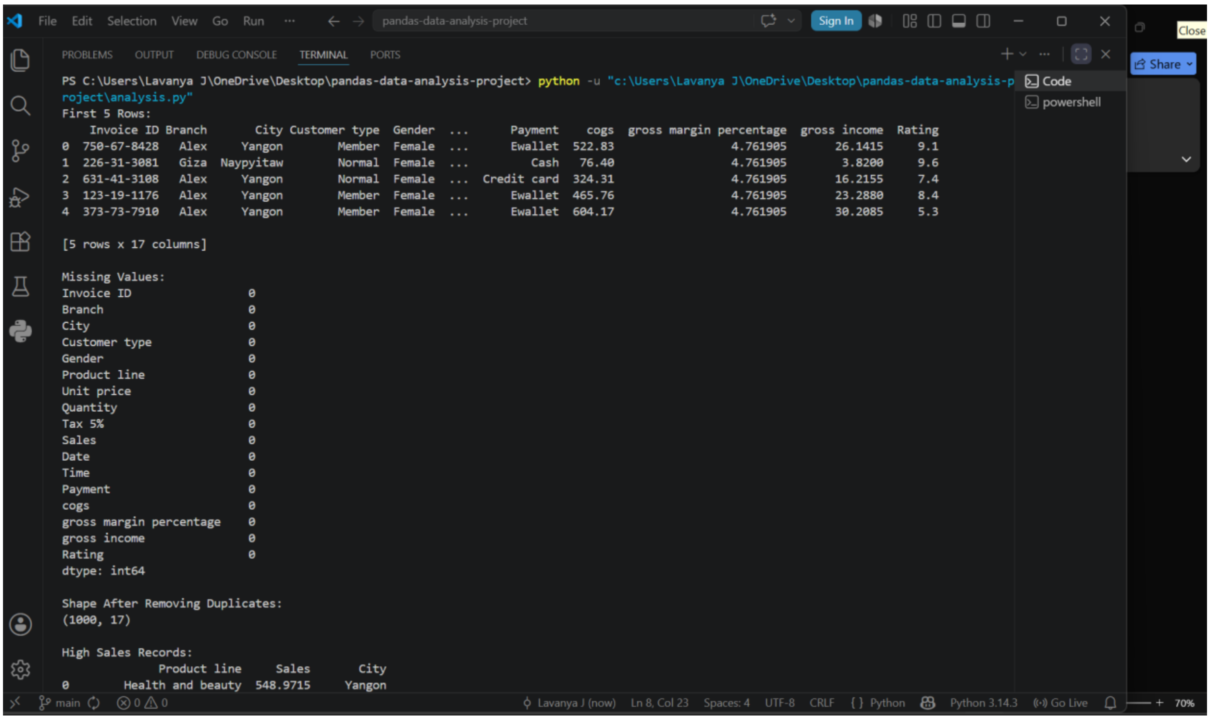


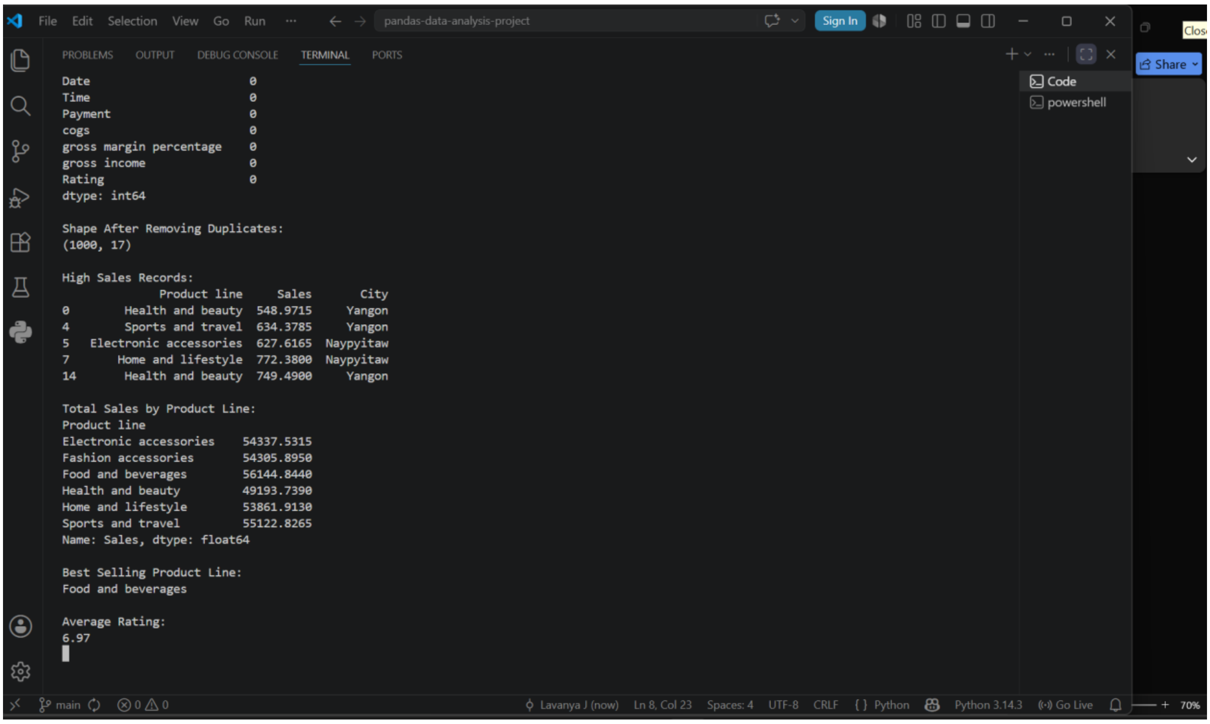
```python
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("dataset.csv")

# Show first 5 rows
print("First 5 Rows:")
print(df.head())

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Remove duplicates
df = df.drop_duplicates()

print("\nShape After Removing Duplicates:")
print(df.shape)

# Filter sales greater than 500
high_sales = df[df["Sales"] > 500]

print("\nHigh Sales Records:")
print(high_sales[["Product line", "Sales", "City"]].head())

# Total sales by product line
product_sales = df.groupby("Product line")["Sales"].sum()

print("\nTotal Sales by Product Line:")
print(product_sales)

# Best selling product line
```

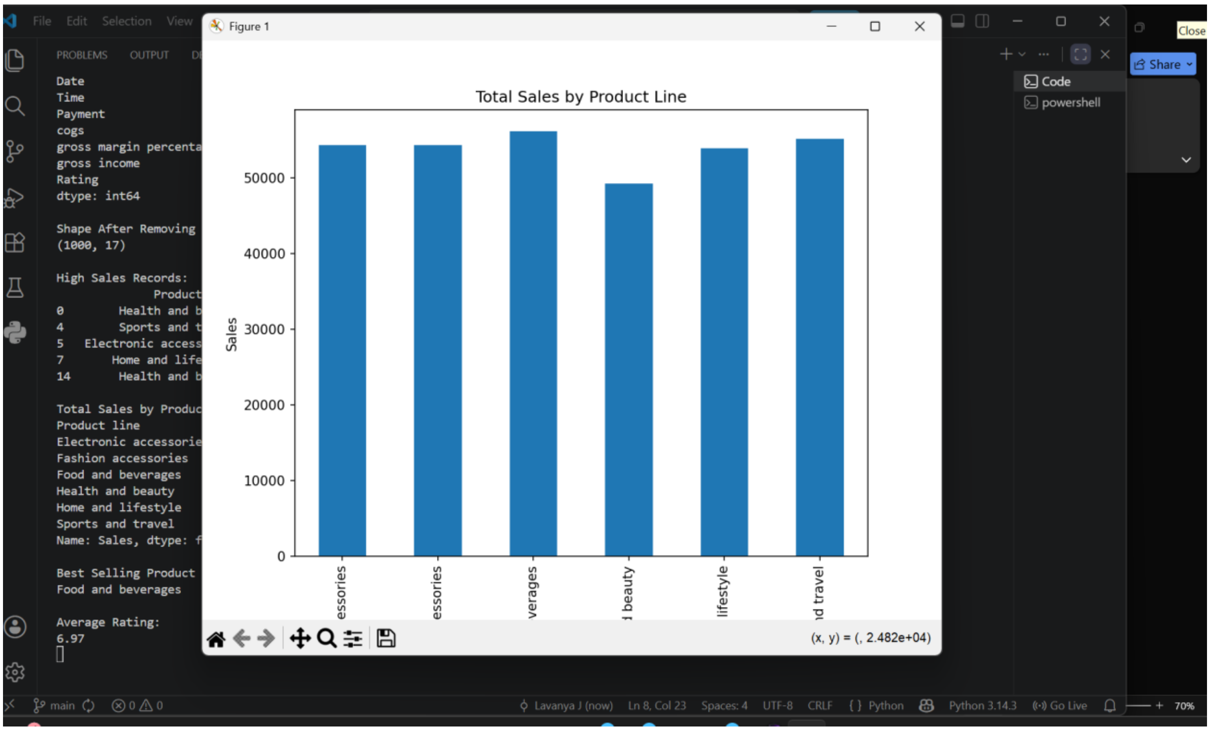


```python

print("\nHigh Sales Records:")
print(high_sales[["Product line", "Sales", "City"]].head())

# Total sales by product line
product_sales = df.groupby("Product line")["Sales"].sum()

print("\nTotal Sales by Product Line:")
print(product_sales)

# Best selling product line
best_product = product_sales.idxmax()

print("\nBest Selling Product Line:")
print(best_product)

# Average rating
average_rating = df["Rating"].mean()

print("\nAverage Rating:")
print(round(average_rating, 2))

# Bar chart for product sales
product_sales.plot(kind="bar")

plt.title("Total Sales by Product Line")
plt.xlabel("Product Line")
plt.ylabel("Sales")

plt.show()
```

Output Screenshots







Insights

1. Product categories generated different levels of sales.
2. High-value sales transactions were identified.
3. Average customer ratings were good.
4. Dataset contained no missing values.

GitHub Repository Link

https://github.com/jlavanya132008-byte/pandas-data-analysis-pro

Hosted Link

https://pandas-data-analysis-project-1.onrender.com

Conclusion

This project helped in understanding practical data analysis using Pandas and visualization using Matplotlib.